

Random Forest Learner

Minitab, LLC has registered RANDOM FORESTS as a trademark, and its usage is authorised. It consists of a predetermined number of decision trees. Each decision tree model is constructed using a unique collection of rows (records), and for each branch,

a random set of columns (describing attributes) is employed. Each decision tree's row sets are generated using bootstrapping and have the same size as the original input table. The attribute set for a decision tree split is determined by randomly selecting $\sqrt{m}$ attributes from the available attributes, where

'm' is the total number of learning columns. The properties may also be supplied in bit (fingerprint), byte, or double vector formats. The output model is applied to the associated predictor node and defines a random forest. This node provides a subset of the Tree Ensemble Learner's capabilities corresponding to a random forest. The configuration window for the Random Forest node that has been chosen for this application can be seen in Figure 7.

Random Forest Predictor

In a random forest model, this function makes pattern predictions based on an aggregation of the predictions made by the individual trees. Figure 8 displays the settings window for the Random Forest Predictor node.

X-Aggregator

This node must be located at the conclusion of a cross-validation loop and must be the node that comes after an X-Partitioner node. It does this by first collecting the result from a predictor node, then comparing the predicted class to the actual class, and finally outputting the predictions for all rows together with the iteration statistics. Figure 9 illustrates the window for the X-Aggregator node's configuration options.

Scorer

This displays the confusion matrix (the number of rows that share an attribute with their classification) by comparing two columns based on attribute value pairs. You may also highlight cells in this matrix to reveal the underlying rows. The comparison dialogue lets you pick two columns to compare; the rows of the confusion matrix reflect the values in the first column you pick, while the columns reflect the values in the second column. The node returns a confusion matrix where each cell indicates

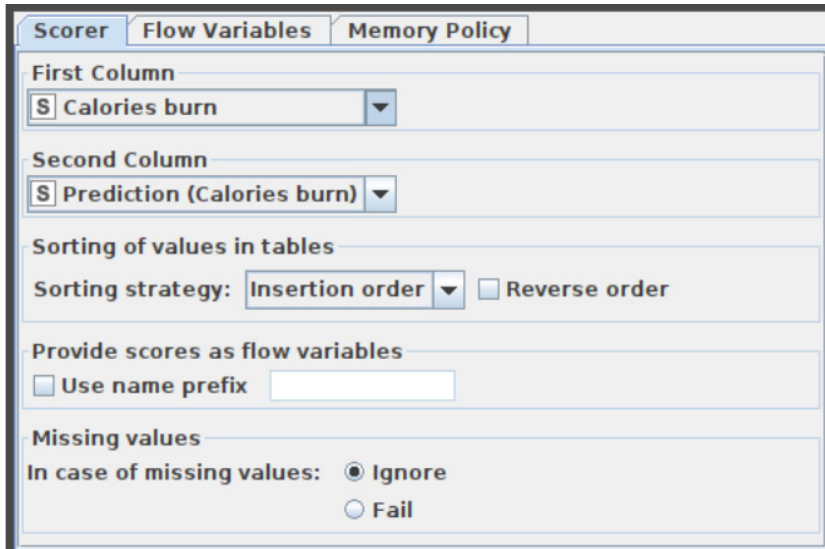


Figure 10: Configuration of Scorer node

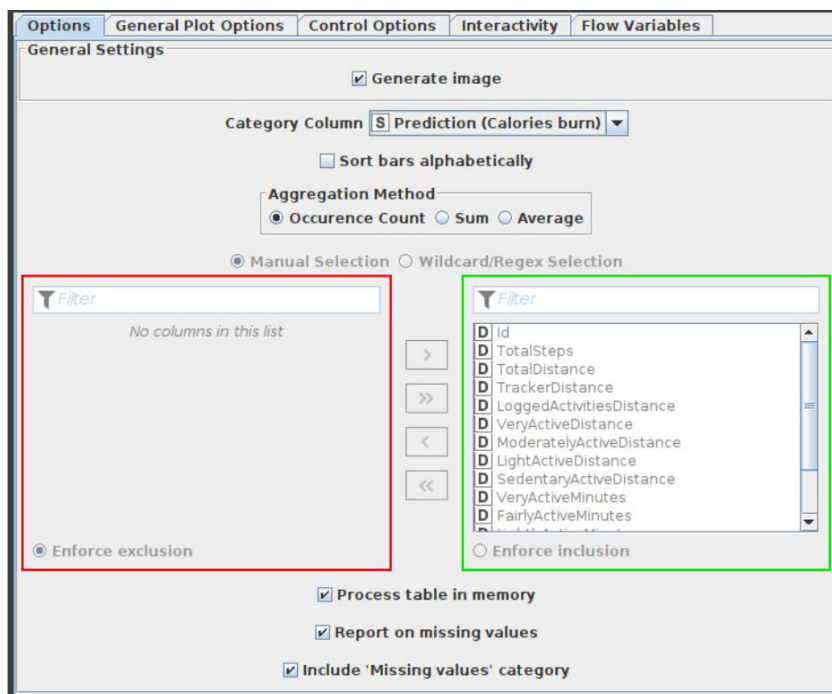


Figure 11: Configuration of Bar Plot node